

## 4 - Dars Report

DATE . . . ( )

### Bank Personal Loan Modelling

1. Kutubxonalar yuklab olindi.

2. Dataset yuklab olindi.

`rows = count, columns = count = data.shape`  
shu kod orqali qatorlar va ustunlarning umumiy sonini bilib oldik.

3. `data.columns` barcha ustunlarni ko'rsatadi.

4. `total = df.isnull().sum().sort_values(ascending=False)`

`percent = (df.isnull().sum() / df.isnull().count()).sort`

`values(ascending=False)`

`missing_data = pd.concat([total, percent], axis=1, keys=`

`['Total', 'Percent'])`.

`return missing_data`

Barcha missing qiymatlarni tekshiruvchi funktsiya yaratib olindi.

5. `missing_check(data)` funktsiyani chaqirish.

6. `data.isnull().sum(), sum()` tekshirib olish

7. Model building

`model_data = data.copy()`

`model_data.drop(['ID', 'ZIP CODE'], axis=1, inplace=True)`

Datasetga o'zgartirish kiritish dan oldin undan nusxa olib qayg'aboldi.

8. Train-test-splitting.



```
X = model_data.drop(['Personal Loan'], axis=1)
y = model_data['Personal Loan']
X_train, X_test, y_train, y_test = train_test_split(X, y,
    test_size=0.3, random_state=42)
```

```
X_train.head()
```

Datani train va testga ajratib oldik  
30% test uchun olindi.

9. LRM = Logistic Regression (solver = 'liblinear')

```
LRM.fit(X_train, y_train)
```

Train qilindi endi test set bilan accuracy  
ni ko'rib olamiz.

```
logistic_test_predict = LRM.predict(X_test)
```

```
LRM_accuracy = metrics.accuracy_score(y_test,
    logistic_test_predict)
```

```
print('Accuracy: {:.4f}'.format(LRM_accuracy))
print('')
```

Accuracy score aniqlab olindi 0.8460 chiqdi.

10. Confusion matrix

```
logistic_cm = metrics.confusion_matrix(y_test,
    logistic_test_predict, labels=[1, 0])
```

```
print(logistic_cm)
```

confusion matrix chiqarildi.

```
[[ 84  65]
```

```
 [ 16 1335]]
```